



University of Al-Ameed
College of Medicine

Department of Biochemistry



glycolysis

Assistant prof.

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11.10.2025

Specific Objectives

1. What is glycolysis? Define. What is meant by aerobic/and anaerobic glycolysis?
2. Study the sequence of reactions involved in glycolytic pathway, enzymes and coenzymes required for each reaction.
3. Differentiate the enzyme 'hexokinase' and 'glucokinase' in phosphorylation of glucose.
4. Note the end-product of glycolytic pathway (a) In presence of O₂ (aerobic) and (b) In absence of O₂ (anaerobic).
5. Learn how regulation of glycolysis is done by substrate, end-products and hormones.
6. Learn the energetics of glycolysis-ATP formation in presence of O₂ and in absence of O₂.
 - What is substrate-level phosphorylation? Give examples from glycolytic pathway.
7. Fate of pyruvic acid formed from glucose
 - (a) If O₂ is available, the pyruvic acid is converted to 2-carbon unit "acetyl-CoA" ("active acetate") by the process of oxidative decarboxylation.
 - (b) If O₂ is not available/or in relative anaerobiosis, pyruvic acid is converted to lactic acid. Learn the details of the reactions involved, enzyme and coenzyme required.

GLYCOLYSIS

- Glycolysis is the *sequence of reactions that converts glucose into pyruvate in the presence of oxygen (aerobic) or lactate in the absence of oxygen (anaerobic) with the production of ATP.*
- Glycolysis is the major pathway for the utilization of glucose and is found in **cytosol** of all cells.

Ist phase: Energy requiring phase or preparative phase

- ✓ 1. Glucose is phosphorylated to glucose-6-phosphate by *hexokinase* or *glucokinase* and ATP is required as a phosphate donor. This is an *irreversible* reaction.
- 2. Conversion of glucose-6-phosphate to fructose-6-phosphate by an enzyme *phosphohexose isomerase* which involve isomerization and is reversible reaction.

- ✓ **3.** Fructose-6-phosphate to fructose 1, 6-bisphosphate, second phosphorylation reaction requiring ATP catalyzed by an enzyme phospho-fructokinase-I. This step is irreversible.
- 4.** Fructose-1,6-bisphosphate is cleaved by aldolase to two three carbon compounds, glyceraldehyde- 3-phosphate and dihydroxy acetone phosphate (DHAP).
- 5.** DHAP is isomerized to glyceraldehyde-3-phosphate by the enzyme phosphotriose isomerase, so that, 2-molecules of glyceraldehyde-3-phosphate are formed from one molecule of glucose.

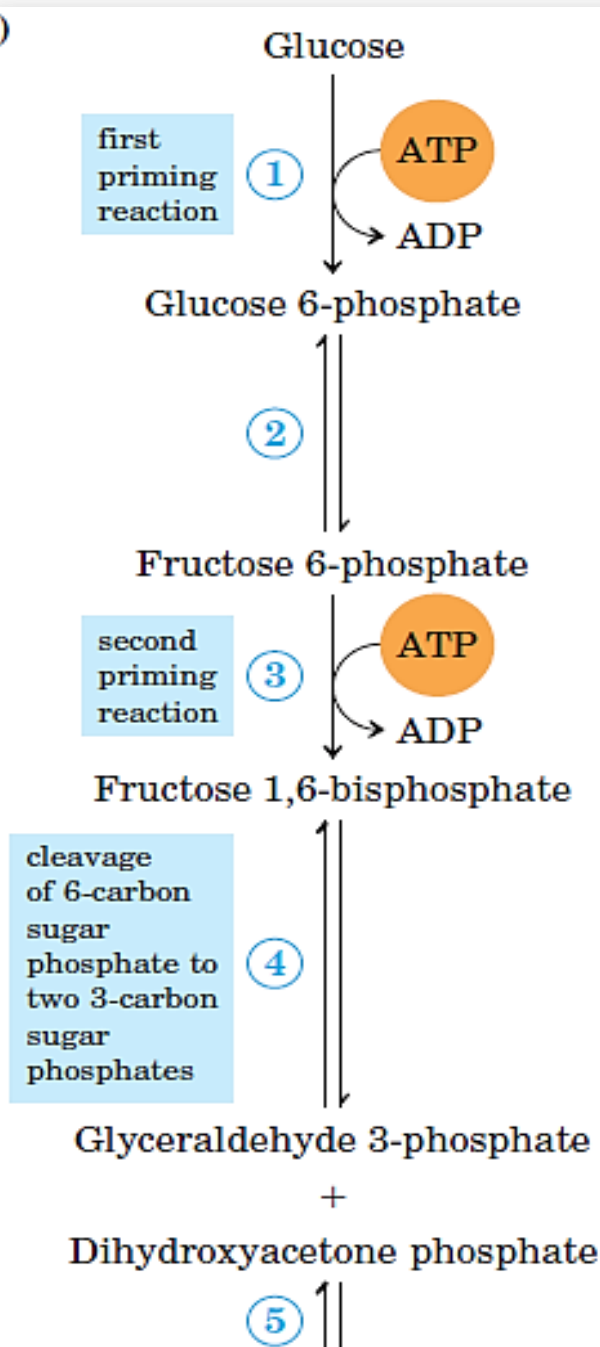
2nd phase: Energy generating phase

- ✓ 6. Oxidation of glyceraldehyde-3-phosphate to 1,3-bisphosphoglycerate by *glyceraldehyde-3-phosphate dehydrogenase*, is a NAD dependent reversible reaction.
- ✓ 7. 1, 3-bisphosphoglycerate to 3-phosphoglycerate is catalyzed by *phosphoglycerate kinase*. This is the step in glycolysis that generates ATP at substrate level phosphorylation. Since two molecules of glyceraldehyde-3-phosphate are formed per molecule of glucose undergoing glycolysis, two molecules of ATP are generated at this stage per molecule of glucose.
- 8. 3-phosphoglycerate to 2-phosphoglycerate is a reversible reaction catalyzed by *phosphoglycerate mutase*.

9. 2-phosphoglycerate to phosphoenolpyruvate. This reaction is catalyzed by *enolase*..

✓ 10. Phosphoenolpyruvate to pyruvate is an irreversible reaction catalyzed by *pyruvate kinase*. This is the second step in glycolysis that generates ATP at substrate level phosphorylation.

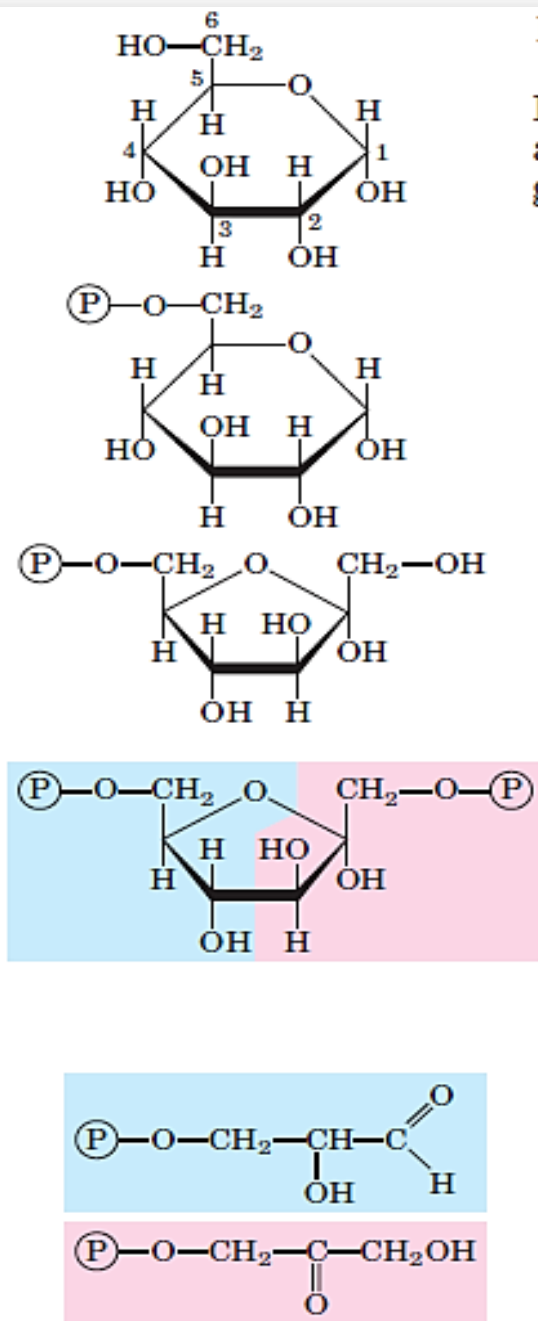
(a)



Preparatory phase

Phosphorylation of glucose and its conversion to glyceraldehyde 3-phosphate

- ① Hexokinase
- ② Phosphohexose isomerase
- ③ Phosphofructokinase-1
- ④ Aldolase
- ⑤ Triose phosphate isomerase



(b)

Payoff phase

Oxidative conversion of glyceraldehyde 3-phosphate to pyruvate and the coupled formation of ATP and NADH

oxidation and phosphorylation

⑥

first ATP-forming reaction (substrate-level phosphorylation)

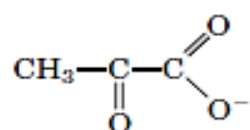
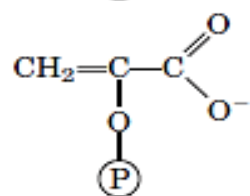
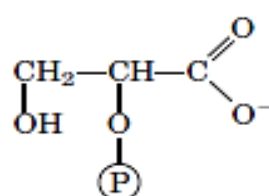
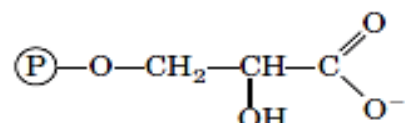
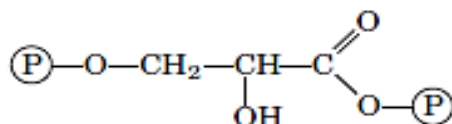
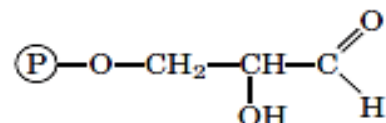
⑦

⑧

⑨

⑩

second ATP-forming reaction (substrate-level phosphorylation)



⑥ Glyceraldehyde 3-phosphate dehydrogenase

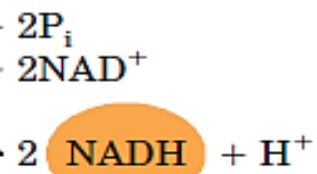
⑦ Phosphoglycerate kinase

⑧ Phosphoglycerate mutase

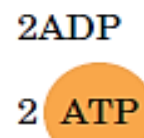
⑨ Enolase

⑩ Pyruvate kinase

Glyceraldehyde 3-phosphate (2)



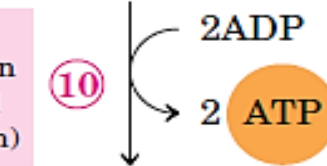
1,3-Bisphosphoglycerate (2)



3-Phosphoglycerate (2)

2-Phosphoglycerate (2)

Phosphoenolpyruvate (2)



Pyruvate (2)

NOTE

- *Under aerobic condition, pyruvate is taken up into mitochondria and after conversion to acetyl-CoA is oxidized to CO_2 and H_2O by citric acid cycle.*
- *In anaerobic conditions, the reoxidation of NADH by respiratory chain is prevented and gets reoxidized by conversion of pyruvate to lactate by **lactate dehydrogenase***
- *Tissues that function under hypoxic conditions produce lactate, e.g. skeletal muscle, smooth muscle and erythrocytes.*
- *In erythrocytes even under aerobic conditions, glycolysis terminates in lactate because of absence of mitochondria.*

Significance of the Glycolysis Pathway

- 1. It is the only pathway that is taking place in all the cells (cytoplasm) of the body.*
- 2. Glycolysis is the only source of energy in erythrocytes.*
- 3. In strenuous exercise, when muscle tissue lacks enough oxygen, anaerobic glycolysis forms the major source of energy for muscles.*
- 4. The glycolytic pathway provides carbon skeletons for synthesis of non-essential amino acids as well as glycerol part of fat (glycerol which can be derived from glucose through dihydroxyacetone phosphate (DHAP))*
- 5. Most of the reactions of the glycolytic pathway are reversible, which are also used for gluconeogenesis.*

Hexokinase

Present in extrahepatic tissue

High affinity for its substrate glucose (low K_m)

Inhibited by its product glucose-6-phosphate

Its function is to ensure supply of glucose for the tissues, irrespective of blood glucose concentration

Catalyze the phosphorylation of other hexoses like fructose, galactose, etc.

Its activity is not affected by insulin

Glucokinase

Present in liver

Low affinity for its substrate glucose (high K_m)

No inhibition by its product glucose-6-phosphate

Its function is to remove glucose from the blood, when the blood glucose level increases (following meal)

Specific for glucose

It is an inducible enzyme that increases its synthesis in response to insulin

Regulation of Glycolysis

Glycolysis is regulated at 3 steps which are irreversible
These reactions are catalyzed by:

1. Hexokinase and glucokinase
2. Phosphofructokinase-I
3. Pyruvate kinase

1- Hexokinase and glucokinase

- ❑ Hexokinase is an allosteric enzyme, that is inhibited by its product glucose-6-phosphate.
- ❑ Liver glucokinase is an inducible enzyme that increases its synthesis in response to insulin and decreases in response to glucagon

2- Phosphofructokinase-1:

- Phosphofructokinase-1 is an inducible enzyme that increases its synthesis in response to *insulin* and decreases in response to *glucagon*
- also Phosphofructokinase-1 is allosteric enzyme *activated by:*
 - *Fructose-6-phosphate (substrate)*
 - *AMP (which signals low energy state)*
 - *Fructose 2,6-bisphosphate.*

And its *inhibited* by *citrate,*
c-AMP and ATP

3- Pyruvate kinase:

- ❖ Pyruvate kinase is an inducible enzyme that increases in concentration with high *insulin* levels and decreases with *glucagon*.
- ❖ It is activated by *fructose-1,6-bisphosphate* and inactivated by *ATP*

Enzyme	Activation	Inhibition
HK		G-6-P
GK	Insulin	Glucagon
PFK	Insulin, AMP F-6-P, PFK-2 F2,6-BP	Glucagon, ATP Citrate, low pH CyclicAMP
PK	Insulin, F1,6-BP	Glucagon, ATP CyclicAMP
PDH	CoA, NAD	Acetyl CoA, NADH

1- What is the first step of glycolysis, and why is it important

2- In which step of glycolysis does each of the following occur?

a. First ATP-consuming reaction

b. Use of NAD as an oxidizing agent

c. First ATP-producing reaction

3- Of the 10 steps of glycolysis, which ones involve phosphorylation

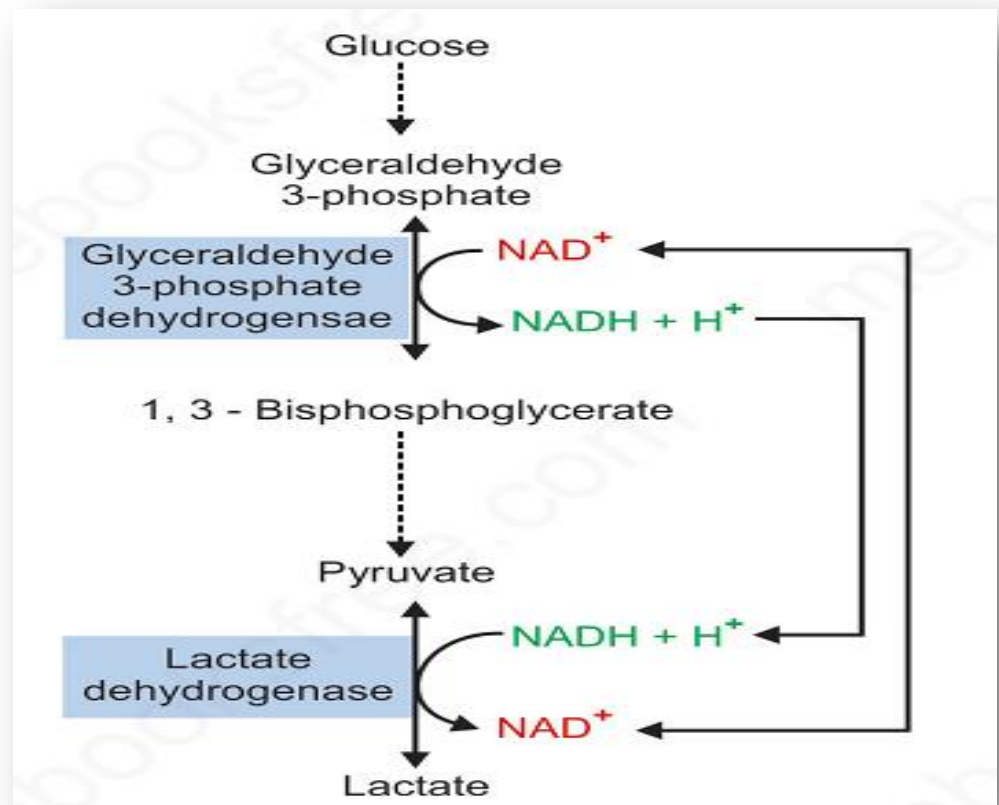
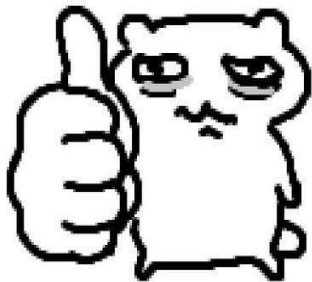


Metabolic fate of pyruvate

1- Reduction of pyruvate to lactate

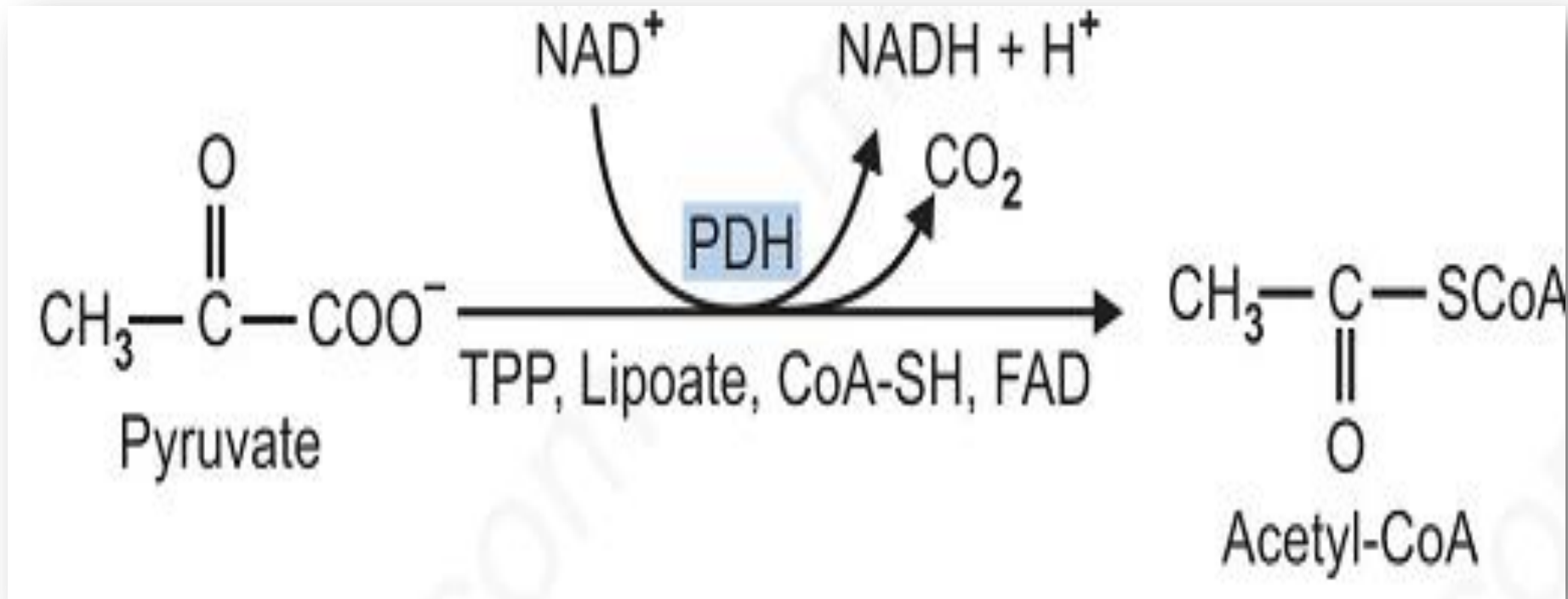
Reduction of pyruvate to lactate by lactate dehydrogenase under anaerobic condition

lets keep going



2- Oxidative decarboxylation of pyruvate

Glycolysis is taking place in cytoplasm. So pyruvate is generated in cytoplasm then transported into mitochondria by a pyruvate transporter. Inside the mitochondria, pyruvate is oxidatively decarboxylated to acetyl CoA by Pyruvate Dehydrogenase (PDH), in tissues with a high oxidative capacity.

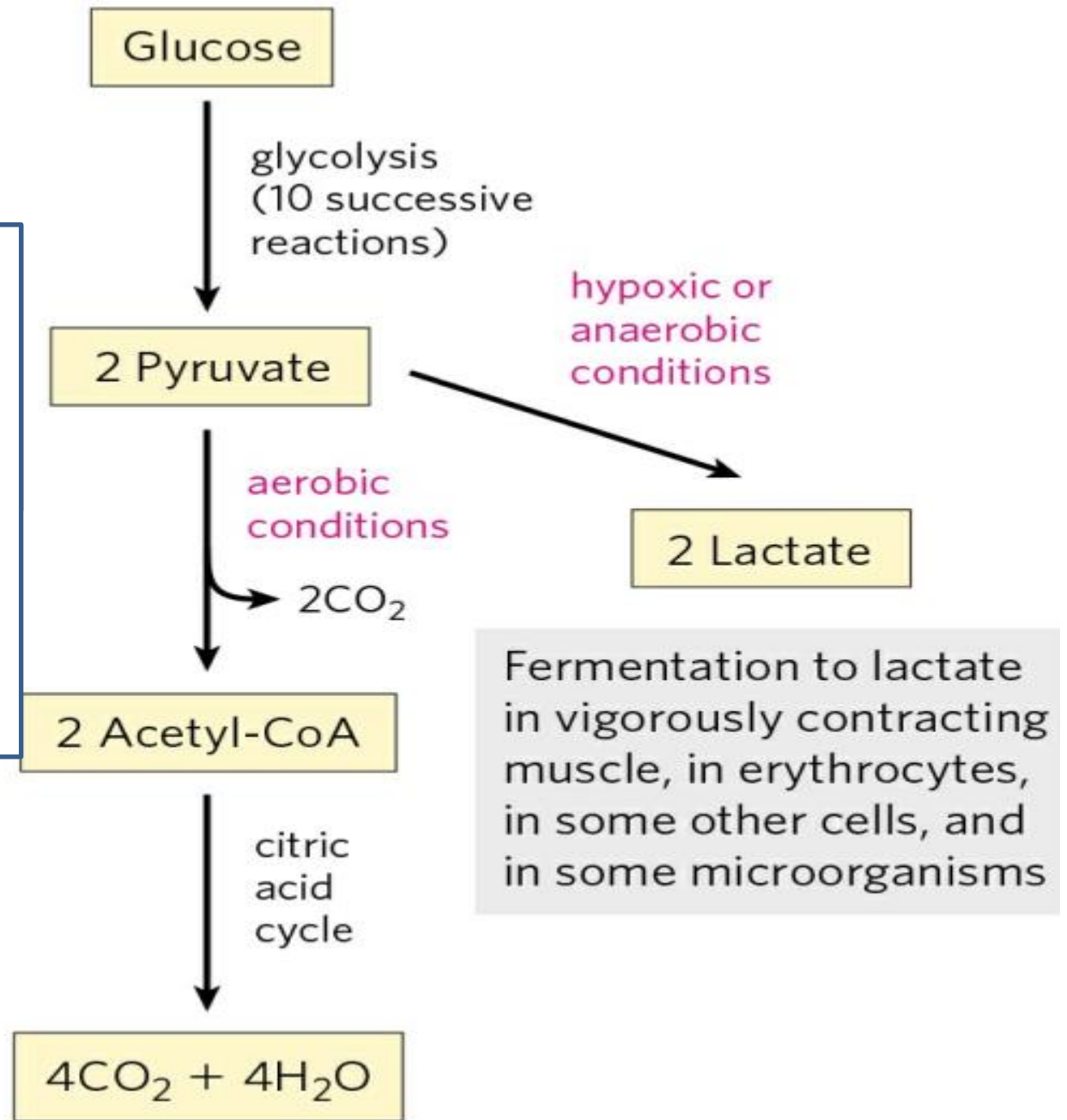


Pyruvate dehydrogenase irreversibly converts pyruvate into acetyl CoA, a major fuel for the tricarboxylic acid cycle .

It is a multi-enzyme complex with 5 co-enzymes, which are:

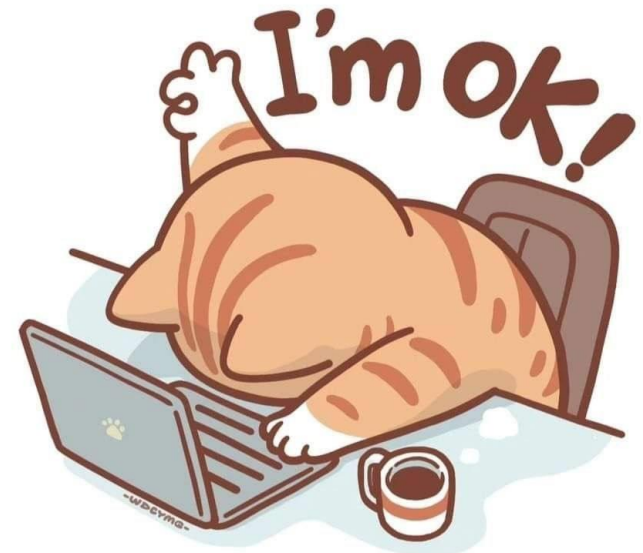
- 1. Thiamine pyrophosphate (TPP)*
- 2. Co-enzyme A (CoA)*
- 3. FAD*
- 4. NAD⁺*
- 5. lipoic acid*

Metabolic fate of pyruvate



Q- In the production of ATP during glucose metabolism in erythrocytes, which of the following is the immediate donor of the phosphoryl group to ADP?

- A.** 2,3-bisphosphoglycerate
- B.** fructose 1,6-bisphosphate
- C.** fructose 2,6-bisphosphate
- D.** glyceraldehyde 3-phosphate
- E.** phosphoenolpyruvate



anaerobic glycolysis

<i>Step</i>	<i>Enzyme</i>	<i>Source</i>	<i>No. of ATPs gained per glucose mol</i>
1	Hexokinase	–	Minus 1
3	Phosphofructokinase	–	Minus 1
6	1,3-bisphosphogly- cerate kinase	ATP	$1 \times 2 = 2$
9	Pyruvate kinase	ATP	$1 \times 2 = 2$
Total = 4 minus 2 = 2			

THANK YOU



aerobic glycolysis

<i>Step</i>	<i>Enzyme</i>	<i>Source</i>	<i>No. of ATPs gained per glucose mol</i>
1	Hexokinase	–	Minus 1
3	Phosphofructokinase	–	Minus 1
5	Glyceraldehyde-3-phosphate dehydrogenase	NADH	$3 \times 2 = 6$
6	1,3-bisphosphoglycerate kinase	ATP	$1 \times 2 = 2$
9	Pyruvate kinase	ATP	$1 \times 2 = 2$
Total = 10 minus 2			= 8

The End!